

On the Convolutions of Sea Wave Spectrum in Radar Backscattering From Ocean Surfaces

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Abstract—By expanding the first-order small-slope approximation (SSA-1) scattering model in the polynomial terms containing convolutions of the sea wave spectrum, this letter quantitatively investigated how the convolutions of the sea wave spectrum affect the normalized backscattering cross section (NBCS) under various wind speeds, radar frequencies, and incident angles. First, numerical results show that only a portion of convolution powers of the sea wave spectrum contribute significantly to the NBCS, and such a portion of convolution powers are defined as the “effective convolutions” (ECs). Based on this, an empirical model related to the radar frequency, incident angle, and wind speed is established to determine the ECs to reduce the computation cost. Finally, the NBCSs using the ECs by the empirical model and the reference data calculated by the direct numerical integration are compared. The comparison results indicate that compared to the direct numerical integration, implementing the first-order SSA with the ECs has high accuracy and relatively low computational cost, notably with small Rayleigh parameters.

Index Terms—First-order small-slope approximation (SSA-1), radar remote sensing, sea surface scattering, sea wave spectrum.

I. INTRODUCTION

IN RADAR remote sensing of the oceans, predicting the normalized backscattering cross section (NBCS) from the sea surface is crucial for ocean hydrodynamic parameter retrieval [1]. Sea surface roughness is multiscale due to the composite of short or capillary waves driven by the local wind, the long waves generated by the air–sea interactions, and the relatively weak nonlinear processes [2]. In the past decades, scattering models have been developed by considering these factors. The small-slope approximation (SSA) [3] has been widely used for various rough terrestrial surfaces, and it works well when the rough surfaces contain a wide range of spectrum components such as sea surface [4], [5], [6]. However, compared to the SSA-1, SSA-2, which has a more complicated

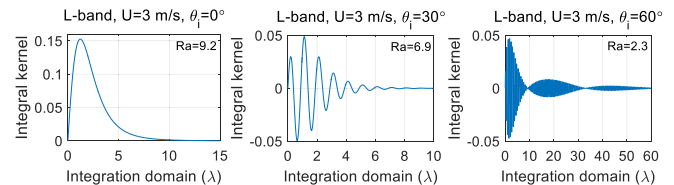


Fig. 1. Integral kernel of the SSA-1 model (see [6, eq. (36)]) as a function of the size of the integration domain (in wavelength) at different incident angles (L-band (1.26 GHz), upwind direction, and a wind speed of 3 m/s; Ra abbreviates the Rayleigh parameter).

mathematical expression in predicting the co-polarized (HH or VV) NBCS, improves the mean accuracy only by 1 dB [6]. Therefore, for predicting co-polarized NBCS, the SSA-1 model is acceptable in terms of accuracy and computation efficiency [7]. This study focuses on the SSA-1 model for co-polarizations.

The literature [4] and [6] computed the NBCS with the SSA-1 through direct numerical integration. The computation efficiency depends on the size of the integration domain and the oscillations of the integral kernel in the SSA-1 model (see [6, eq. (36)]). However, as shown in Fig. 1, when the incident angle increases, the direct numerical integration suffers from a large integration domain size at the L-band and a low wind speed of 3 m/s (i.e., at smaller Rayleigh parameters). Meanwhile, a finer discretization is required for which the integral kernel is highly oscillatory at small Rayleigh parameters. Besides, it has been found that the computation efficiency can be increased when the exponential term in SSA-1 is approximated by the summation of finite Taylor series expansion terms [8]. The SSA-1 model may be expressed as a summation of the sea wave spectrums at different orders instead of an integral expression in [4] and [6]. Also, the literature [8] discussed the upper limit of these expansion terms and suggested 150 or more expansion terms to ensure convergence for the case of L-band and low wind speed of smaller than 3 m/s. However, there is no specific expression that aggregates all three variables of radar frequency, wind speed, and incident angle, or the Rayleigh parameter for selecting the upper limit (or perhaps a lower one) in the series expansion. For large Rayleigh parameters, though the literature [8] proposed to use the numerical integration implementation in [4], it is still not easy to compute the NBCS under general observation configurations if a hybrid of the series expansion with direct numerical integration is applied.

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This letter seeks the specific Taylor series terms in the SSA-1 model, which significantly contribute to the NBCS. Specifically, how the convolutions of the sea wave spectrum affect the radar backscattering from ocean surfaces is revealed. Meanwhile, the effective series expansion terms with significant contributions to the NBCS under general radar probing frequencies (incidence angles and wind speeds) are further determined, which help to improve the computation efficiency. Based on this, the SSA-1 numerical computation can be realized more efficiently and its application scope can be expanded under general observation conditions.

II. FUNDAMENTALS AND METHODOLOGIES

A. Expression of SSA-1 With Series Expansion

The general expression of NBCS (without the coherent components) in the SSA-1 model with the series expansion is given as [8]

$$\sigma_{pq}^0(\mathbf{k}_i, \mathbf{k}_s) = 4\pi \cdot \left| \frac{2q_i q_s B_{pq}(\mathbf{k}_i, \mathbf{k}_s)}{Q} \right|^2 \cdot \exp(-\sigma_z^2 Q^2) \cdot \sum_{n=1}^{\infty} \frac{(\sigma_z^2 Q^2)^n}{n!} \cdot W^{(n)}(\mathbf{k}_s - \mathbf{k}_i). \quad (1)$$

With

$$W^{(n)}(\mathbf{k}_s - \mathbf{k}_i) = \frac{1}{4\pi^2} \int_0^\infty \int_0^{2\pi} \rho^n(\mathbf{r}) \cdot \exp[j(\mathbf{k}_s - \mathbf{k}_i) \cdot \mathbf{r}] d\mathbf{r} \quad (2)$$

where $W^{(n)}$ is the Fourier transform of the n th power of the normalized autocorrelation function (ACF) $\rho(\mathbf{r})$ of the sea surface; p and q represent the received and transmitted polarizations, respectively; $\{\mathbf{k}_i, \mathbf{k}_s\}$ are the horizontal projections of the incident and scattered wave vectors, and $\{q_i, q_s\}$ are the absolute values of their vertical projections. $Q = q_i + q_s$, and for the backscattering considered in this study, $Q = 2k \cos \theta_i$, where k is the wavenumber of the radar wave and θ_i is the incident angle. $B_{pq}(\mathbf{k}_i, \mathbf{k}_s)$ is the matrix obtained by the first-order small perturbation method, with the explicit expression found in [5]. For more details on the series expansion and the definitions of other quantities in the classical SSA-1 model, refer to [5], [6], and [8].

B. Sea Wave Spectrum

The sea wave spectrum, which characterizes the sea surface roughness in terms of energy, is commonly used in scattering models. The two-dimensional (2-D) directional sea spectrum can be expressed as the product of the omnidirectional spectrum $S(K)$ and the angular spreading function (ASF) $\Phi(K, \phi)$ [9], and it has the general form in the polar coordinate as

$$W(K, \phi) = \frac{1}{K} S(K) \cdot \Phi(K, \phi) \quad (3)$$

where ϕ is the azimuth angle relative to the upwind direction and K denotes the sea wave wavenumber. The variance σ_z^2 and the normalized ACF $\rho(\mathbf{r})$ of the sea surface elevation in the polar coordinate are directly bridged to $S(K)$ and $W(K, \phi)$

by the Wiener–Khinchin theorem, namely

$$\sigma_z^2 = \int_0^\infty S(K) dK \quad (4)$$

$$\rho(r, \phi) = \frac{1}{\sigma_z^2} \int_0^\infty \int_0^{2\pi} W(K, \phi) \cdot \exp(jKr \cos(\phi - \phi)) K dK d\phi. \quad (5)$$

According to the convolution property, (2) is related to the $(n-1)$ -fold convolution of the sea wave spectrum $W(K, \phi)$ as

$$W^{(n)}(K, \phi) = \underbrace{W(K, \phi) \otimes W(K, \phi) \cdots \otimes W(K, \phi)}_n / (\sigma_z^2)^n \quad (n \geq 1) \quad (6)$$

where $\otimes$ is the convolution operator. Since (6) is defined as the convolution power n of $W(K, \phi)$, the convolution powers evaluated at a specific surface wave component K (i.e., at the Bragg wavenumber and in the backscattering, $K = 2k \sin \theta_i$) along the azimuth of the radar look direction ϕ (relative to upwind direction) contribute to the NBCS [2]. In the following numerical simulations, the newly developed omnidirectional sea wave spectrum $S(K)$ proposed by Hwang [10] and $\Phi(K, \phi)$ in the Elfouhaily spectrum [9] are applied, and a fully developed sea with an inverse wave age at 0.84 is considered.

III. NUMERICAL RESULTS AND DISCUSSION

A. Analysis of Numerical Results

This section examines the behavior of $W^{(n)}(K, \phi)$, the corresponding individual term $\sigma_{(n)}^0$ in the NBCS given in (1), and their cumulative contributions to the NBCS $\sum \sigma_{(n)}^0$, as a function of the series index n at different incidence angles, wind speeds, and radar frequencies. Meanwhile, the Debye equation [11] is applied to obtain the seawater permittivity with fixed sea surface temperature and salinity at 20 °C and 35 ppt, respectively. Fig. 2 shows the numerical results at the L-band (1.26 GHz) and along the upwind direction ($\phi = 0^\circ$), with a wind speed of 5 m/s and incident angles at 0° , 30° , and 60° .

At $\theta_i = 0^\circ$, the Bragg wavenumber K is 0 rad/m, and in Fig. 2(a1), the corresponding $W^{(n)}(K, \phi)$ has a single peak at $n = 2$ and monotonically decreases with the growth of n . However, as shown in Fig. 2(b1), a different trend is observed for $\sigma_{(n)}^0$. It can be reasonably argued that the value of $\sigma_{(n)}^0$ implicitly given in (1) is determined by both $W^{(n)}(K, \phi)$ and $(\sigma_z^2 Q^2)^n / n!$. The variations of $\sigma_{(n)}^0$ with n indicate that only a portion of convolution powers of the sea wave spectrum contribute considerably to the NBCS and can be defined as the “effective convolution” (EC). Moreover, the EC powers do not begin with $n = 1$, which is a new phenomenon compared to that discussed in [8], where only the upper limit of n was considered. In practice, the lower limit is more crucial in reducing the computation cost than the upper one because the calculations may be stopped at a preset threshold. These EC powers are almost symmetrically distributed on both sides of n , where $\sigma_{(n)}^0$ reaches its maximum. Hereafter, such an index n is defined as $n_{\max}$. Outside of the range of ECs, $W^{(n)}$

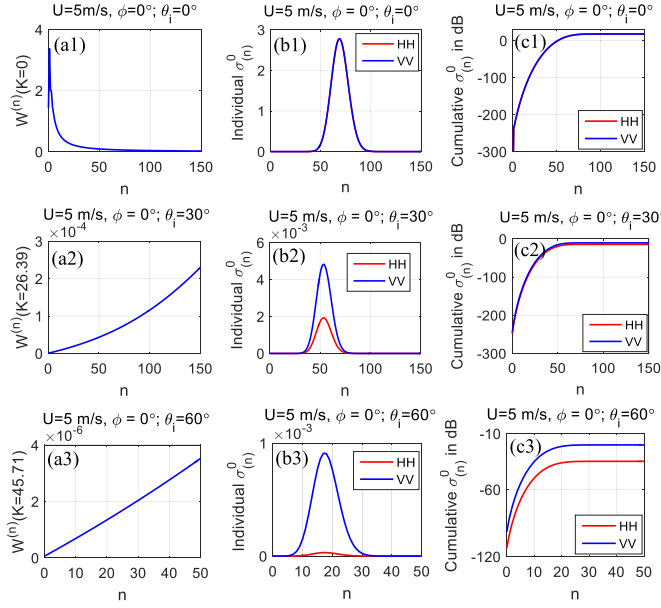


Fig. 2. (a1)–(a3) Convolution powers (in a natural unit), (b1)–(b3) individual contribution to NBCS (in a natural unit), and (c1)–(c3) cumulative contribution to NBCS (in dB) at the L-band (1.26 GHz) and the incidence angles of 0° , 30° , and 60° , respectively, along the upwind direction when the wind speed is 5 m/s.

has negligible contributions to the NBCS. In Fig. 2(c1), the cumulative contributions to the NBCS converge at a specific convolution power n , beyond which the NBCS saturates.

When the incident angle is medium and large, the Rayleigh parameter is small. It can be observed from Fig. 2(a2) and (a3) that $W^{(n)}$ increases monotonically with the growth of n , while its quantitative level decreases with the increment of θ_i . The individual contributions to the NBCS are similar to those at $\theta_i = 0$, but the peak point $n_{\max}$ shifts toward a smaller n , notably when θ_i is large. Due to the left shifts of $n_{\max}$ with the increase in θ_i , the cumulative contributions to the NBCS at a large θ_i converge faster than that at a small θ_i . In the downwind direction, the SSA-1 model with a simple spectral representation for the surface cannot be distinguished from that in the upwind direction since they both have the same simulated NBCSs; meanwhile, for other wind directions like the crosswind, the simulated results are found to have the same trend with that in the upwind direction, but with smaller quantitative levels. In the following, the results in the upwind direction and the nadir incidence at different frequencies and wind speeds are considered.

For the case of the nadir incidence and at the L-band, Fig. 3 shows the individual and cumulative contributions of $W^{(n)}$ to the NBCS at different wind speeds. It can be observed that as the wind speed increases (i.e., the Rayleigh parameter becomes large), $n_{\max}$ shifts to a larger n , and the individual contribution to the NBCS decreases. Consequently, for a higher wind speed, the cumulative contribution to NBCS converges at a larger n . Fig. 4 shows the results calculated at a high wind speed of 15 m/s but for different radar frequencies at the L-band (1.26 GHz), C-band (5.3 GHz), and Ku-band (14 GHz). It can be seen that the peak point $n_{\max}$ shifts to a larger value with increasing frequency (i.e., the Rayleigh

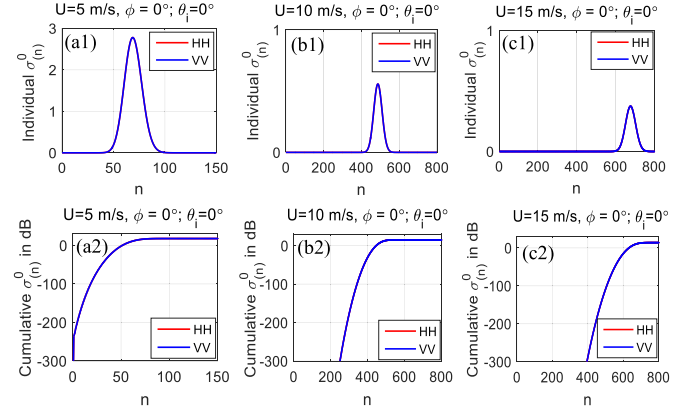


Fig. 3. (a1)–(c1) Individual contribution (in natural unit) and (a2)–(c2) cumulative contribution (in dB) to the NBCS along an upwind direction at L-band (1.26 GHz) with an incidence angle of 0° for wind speeds of 5, 10, and 15 m/s, respectively.

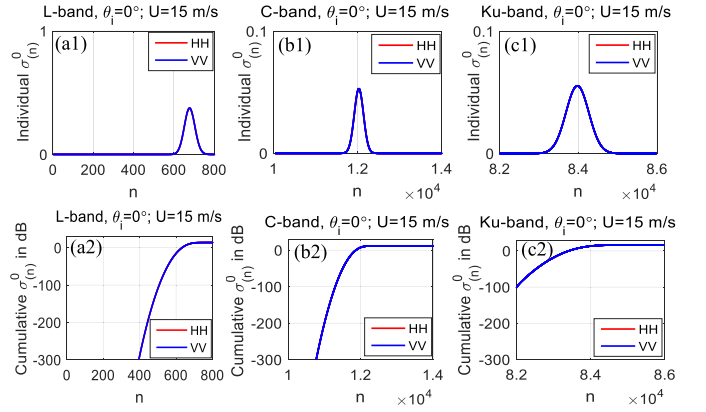


Fig. 4. (a1)–(c1) Individual contribution (in natural unit) and (a2)–(c2) cumulative contribution (in dB) to the NBCS along an upwind direction with an incidence angle of 0° for a wind speed of 15 m/s at L-band (1.26 GHz), C-band (5.3 GHz), and Ku-band (14 GHz), respectively.

parameter becomes large). The results indicate that at a higher frequency (or at a larger Rayleigh parameter), a larger upper limit n is required for the cumulative contribution to NBCS to reach convergence, as shown in Fig. 4(a2)–(c2).

To sum up, this section shows that only a portion of the convolution powers of the sea wave spectrum contributes significantly to the NBCS. The results indicate that the EC powers do not always begin with $n = 1$. It indicates that only giving the upper limit, e.g., $n = 150$ or a larger value in [8], might still not be efficient when applying the SSA-1 model with series expansions, notably at high wind speeds and radar probing frequencies. If the indices of ECs can be determined, the computational efficiency will be significantly improved. To achieve this goal, the issue of quantifying the ECs is investigated in Section III-B.

B. Quantifying the Range of the ECs

Based on the variations of σ_n^0 shown in the last subsection, it is known that the ECs are symmetrically distributed around $n_{\max}$, and such a feature is the key to determining the ECs. Thus, the location of $n_{\max}$ should be determined first. It can

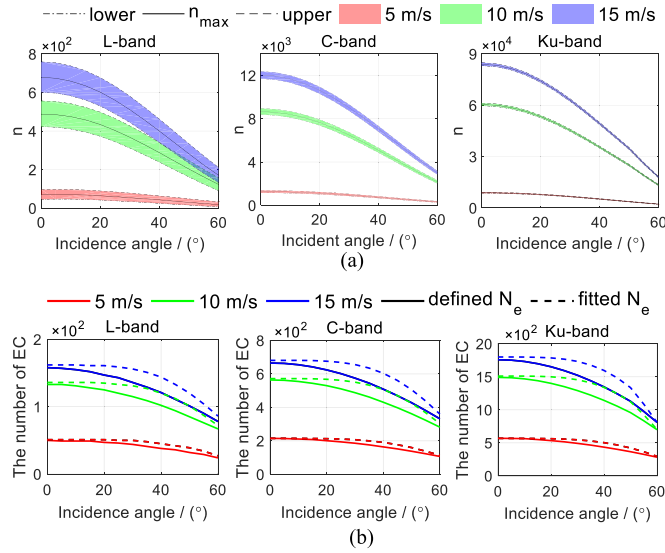


Fig. 5. (a) Lower and upper limits of the defined ECs obtained from the criterion in (9) and (b) defined N_e , i.e., the difference between the lower and upper limits shown in Fig. 5(a) and the fitted N_e calculated by (12) versus the incidence angles for the wind speeds of 5, 10, and 15 m/s at the L-, C-, and Ku-bands, respectively.

be observed from (1) that the index n_{max} can be obtained from the function $g(n) = (\sigma_z^2 Q^2)^n W^{(n)}/n!$ when $g'(n) = 0$. By regarding it as a continuous function of x , we have

$$g'(x) = \frac{(\sigma_z^2 Q^2)^x}{x!} W(x) \cdot \left[\ln(\sigma_z^2 Q^2) - \varphi(x+1) + \frac{W'(x)}{W(x)} \right] \quad (7)$$

where $\varphi(x)$ is the Digamma function; the item $W'(x)/W(x)$ can be ignored due to the negligible increment of $W^{(n)}$ compared to that of the index n [see Fig. 2(a)]. Then, by combining the properties of the Digamma function and the Euler–Maschenori constant at a large n , the peak point is found to be at around $\sigma_z^2 Q^2$, which is precisely the expression of the Rayleigh parameter, namely

$$n_{\text{max}} \approx \text{round}(\sigma_z^2 Q^2) \quad (8)$$

where $\text{round}(\cdot)$ is the function for obtaining an integer.

Through checking the cases presented in Figs. 2–4, it is found that the error of the estimated n_{max} by using (8) is within 2, and such an approximation is acceptable. Once the value n_{max} is obtained, the next step is to determine the limit (or boundary) of ECs. Here, the ECs are defined by the criterion that the individual contribution of $W^{(n)}$ to the NBCS is greater than 1% of the maximum one, namely

$$\sigma_{(n)}^0 \geq \sigma_{(n_{\text{max}})}^0 / 100. \quad (9)$$

The convolution powers $W^{(n)}$ satisfying the criterion in (9) are recognized as the “defined ECs.” By adopting this criterion, the lower and upper limits of ECs, i.e., n_{lower} and n_{upper} , at different radar frequencies, incidence angles, and wind speeds are obtained and shown in Fig. 5(a), where the solid line denotes n_{max} . It can be seen that n_{max} and the upper and lower limits of the defined ECs are all associated with

the incident angle, the wind speed, and the radar frequency. To obtain the lower and upper limits of the ECs prior to numerically implementing the SSA-1 model, this study further introduces a parameter, i.e., the number of ECs N_e , which equals the difference between the upper limit of the defined ECs and its lower limit. Since the ECs are symmetry to n_{max} , the lower limit n_{lower} ($n_{\text{lower}} \geq 1$) and the upper limit n_{upper} of ECs can be represented as

$$\begin{cases} n_{\text{lower}} = n_{\text{max}} - N_e/2 \\ n_{\text{upper}} = n_{\text{max}} + N_e/2. \end{cases} \quad (10)$$

The values of N_e calculated based on Fig. 5(a) are plotted with the solid lines in Fig. 5(b). Thus, if the value of N_e is known in advance, the upper and lower limits can be obtained. Therefore, it is crucial to establish a relationship between N_e and radar parameters and wind speed. To this end, the variations of N_e with the radar parameters and wind speeds need to be quantified. The next step is to obtain a specific expression for the number of ECs based on the numerical results shown in Fig. 5(a). According to Fig. 5(a), n_{max}/N_e decreases monotonically with the growth of the incident angle, and it increases monotonically with the increasing wind speed and radar frequency. By using the separation of variables, we have

$$n_{\text{max}}/N_e = u(U, f, \theta) = A(U)B(f)/C(\theta) \quad (11)$$

where the functions $A(U)$, $B(f)$, and $C(\theta)$ increase monotonically with the growth of U , f , and θ , respectively. Meanwhile, (11) indicates that a specific expression of u cannot be uniquely determined. Here, one approach to obtain a specific expression is introduced as follows.

From Fig. 5(a), it can be seen that the relationship between u and f is almost linear such that $B(f) \approx f$. Though the nonlinearity between u and both U and θ is more potent, $A(U) \approx U$ can be still used by letting the term $C(\theta)$ convey the nonlinearity in u . As for $C(\theta)$, it is further treated as a function of $\sin \theta$ instead of θ , which helps to achieve a more concise expression because it avoids the appearance of higher order polynomials of θ and $\sin \theta$ is a nonlinear function. Then, $C(t)$ with $t = \sin \theta$ may have concise forms and can be obtained through curve fittings. Meanwhile, it is desired that the ECs predicted by the fit curves to be a little larger than the defined ECs to maintain a higher accuracy. Together, one approximate expression of $C(t)$ is tuned out to be $4.5 \exp(t^2)$. Finally, the specific expression of ECs is deduced as

$$N_e = \text{round} \left(\frac{4.5 n_{\text{max}} \cdot \exp[\sin^2 \theta]}{U \cdot f} \right). \quad (12)$$

For more intuitive relationships, n_{max} can be replaced by (8) and with $\sigma_z^2 = 4.004 \times 10^{-5} U_{10}^{4.034}$ obtained from the Hwang spectrum for fully developed sea surfaces. N_e calculated by (12) is labeled the “fitted N_e ” and is also shown in Fig. 5(b) with the dotted line.

C. Comparison With Classical SSA-1

To verify the accuracy and efficiency of the proposed method, the NBCS calculated by using the convolution powers

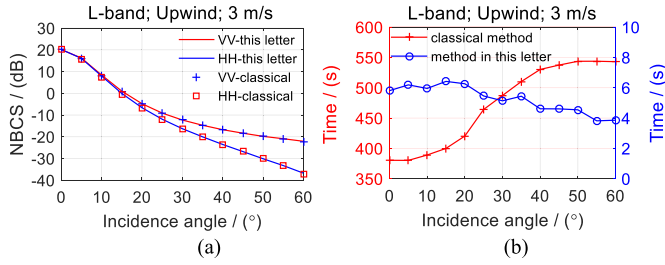


Fig. 6. (a) Accuracy and (b) efficiency comparisons of the SSA-1 with ECs and the classical SSA-1 with the numerical integration at the L-band and a wind speed of 3 m/s.

determined by (12) is compared with that by direct numerical integration [6]. Fig. 6 presents a comparison at the L-band with a wind speed of 3 m/s as an example for these probing parameters corresponding to the small Rayleigh parameters shown in Fig. 1. Meanwhile, the sampling interval is set to $\lambda/350$ in the space domain, and the difference between two adjacent cumulative sums (NBSCs) of less than 10^{-7} is taken as the gauge to ensure the convergence in numerical integration; in the proposed method, the sampling points of 2048×2048 with a sampling interval at around 0.03 rad/m are set to discretize the sea wave spectrum. The numerical computations are conducted on a computer equipped with an Intel¹ Core² i9-10885H CPU @ 2.40 GHz.

Fig. 6(a) shows that the results of the proposed method are highly consistent with that of the classical SSA-1 with the direct numerical integration, indicating that the proposed method is reliable in terms of accuracy. When it comes to the computation time, Fig. 6(b) shows that, at very small Rayleigh parameters, the time cost of the proposed method is in the order of a few seconds, indicating that the method is much more efficient than the SSA-1 with the direct numerical integration. The improvement in computation efficiency is more obvious at larger incident angles. The proposed method provides an alternative approach for the accurate and effective implementation of the SSA-1 model.

IV. CONCLUSION

By expanding the SSA-1 model into a series expression in terms of the sea wave spectrum, this letter quantitatively investigated the variations of the convolution powers of the sea wave spectrum and their individual and cumulative contributions

to the NBSC as a function of the index n . First, the numerical results provide deep insights into the effects of the convolutions of sea wave spectrum on radar backscattering. Based on these results, the concept of ECs was proposed, i.e., the convolutions of the sea wave spectrum with significant contributions to the NBSC do not always begin with $n = 1$, and only a portion of sea spectrum convolutions are effective. Then, an empirical relationship between the number of ECs and the observation parameters (radar frequency, incident angle, and wind speed) was further established. With this relationship, the SSA-1 model can be implemented with the series expansion expression under general observation conditions to achieve the desired computational accuracy and efficiency.

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